

POSTNIKOV TRUNCATIONS AS JOINS OF SHEAF AND HOMOTOPY EQUIVALENCES

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ABSTRACT. In the complete lattice $\text{Same}(\mathbf{sSet})$ of all two-out-of-three classes of morphisms of simplicial sets, let W_{test} be the weak homotopy equivalences and W_{D_k} the maps inverted by the k -coskeleton, i.e. the sheaf equivalences of the k -th coskeletal subtopos. We prove that for every $n \geq -1$ the join $W_{\text{test}} \vee W_{D_{n+1}}$ is exactly the class of n -truncated weak equivalences: the Postnikov truncation is the lattice join of sheaf sameness and homotopy sameness, with no fibrancy hypothesis, and $W_{\text{test}} = \bigcap_n (W_{\text{test}} \vee W_{D_n})$ (a sheaf-theoretic Whitehead theorem). The proof is an instance of a general join theorem: if an endofunctor P is connected to the identity by a natural transformation with components in $U \vee V$ and sends U and V into a sameness $W \subseteq U \vee V$, then $U \vee V = P^*(W)$; for two reflectors L_1, L_2 this gives $W_{L_1} \vee W_{L_2} = W_{L_2 L_1}$ iff $L_2 L_1$ inverts W_{L_2} . To set this up we show that for any Grothendieck topology J on a small category S the class W_J of maps inverted by sheafification is an element of $\text{Same}(\text{Presh}(S))$, that $J \mapsto W_J$ is an order-embedding whose image consists of the saturated samenesses inverted by a left-exact reflector, and that, within the embedded sheaf samenesses, joins in the lattice of topologies are obtained by saturating joins in the sameness lattice: $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$. We also give a short direct proof, in this language, of the finite-sieve form of the Kelly–Lawvere correspondence — the Grothendieck topologies on a category with finitely many sieves on each object are its idempotent two-sided ideals — and re-derive from it the classical facts that the subtoposes of simplicial and cubical sets are the coskeletal ones; the dictionary is used to locate the sheaf chain W_{D_k} inside $\text{Same}(\mathbf{sSet})$. All finite claims are verified by exhaustive computation from the definitions.

1. INTRODUCTION

Two theories may agree on what an object *is* and disagree on when two objects are *the same*. Homotopy theory identifies simplicial sets along weak equivalences; sheaf theory identifies presheaves that have the same sheafification. Both notions of sameness are classes of morphisms satisfying two-out-of-three, and the companion papers [1, 2] proposed to study all such classes

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at once: for a category C , a *sameness* is a class W of morphisms containing the identities, closed under composition and satisfying two-out-of-three, and $\text{Same}(C)$ is the complete lattice of samenesses under inclusion. The present paper compares the sheaf-theoretic and the homotopical samenesses of simplicial sets *inside* this lattice.

Main theorem. Let $W_{\text{test}} \in \text{Same}(\mathbf{sSet})$ be the weak homotopy equivalences and, for $k \geq 0$, let W_{D_k} be the class of maps f with $\text{cosk}_k f$ an isomorphism — equivalently, bijective in dimensions $\leq k$ — which is the class of maps inverted by sheafification for the k -th coskeletal subtopos. For $n \geq -1$ let T_n be the n -truncated weak equivalences (maps inducing isomorphisms on π_i for $i \leq n$ at all base points; for $n = -1$, maps whose source and target are both empty or both non-empty). Then (Theorem 7.2 and Corollary 7.3)

$$W_{\text{test}} \vee W_{D_{n+1}} = T_n \quad (n \geq -1),$$

$$W_{\text{test}} = \bigcap_{n \geq 0} (W_{\text{test}} \vee W_{D_n}), \quad \text{Iso} = \bigcap_{n \geq 0} W_{D_n}.$$

That the coskeleta of a *Kan complex* model its Postnikov sections is classical; the point here is the statement about *classes of maps*: the Postnikov truncation, which at the ∞ -level is the n -truncation modality, is the lattice join of the sheaf sameness of the coskeletal topology with the homotopy sameness, on all simplicial sets, and the homotopy sameness is the meet of these joins. In words,

$$\text{sheaf localization} \vee \text{homotopy localization} = \text{Postnikov localization}.$$

The general join theorem. The proof is an instance of a statement about arbitrary samenesses (Theorem 4.1): if an endofunctor P of C receives a natural transformation from the identity whose components lie in $U \vee V$, and P sends U and V into a sameness $W \subseteq U \vee V$, then $U \vee V = P^*(W)$, the class of maps sent by P into W . For two reflectors L_1, L_2 onto full subcategories, with W_{L_i} the maps they invert, this specialises to $W_{L_1} \vee W_{L_2} = W_{L_2 L_1}$ iff $L_2 L_1$ inverts W_{L_2} (Corollary 4.2); for two Grothendieck topologies J, J' it shows that $W_{J \vee J'}$ is the saturation of $W_J \vee W_{J'}$ (Proposition 4.3). The main theorem is the case $U = W_{\text{test}}, V = W_{D_{n+1}}, P = \text{cosk}_{n+1} \circ \text{Ex}^\infty, W = W_{\text{test}}$.

Sheaf sameness. To place sheaf theory in the lattice we show (Theorem 3.1) that for every Grothendieck topology J on a small category S the class W_J of maps inverted by sheafification is a sameness, that $J \mapsto W_J$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$, that W_J is the class of J -local isomorphisms, and that the image consists exactly of the *saturated* samenesses that are the full class of maps inverted by a left-exact reflector. Saturation is essential: $\{\text{id}\}$ and Iso are distinct samenesses with the same (identity) localization, and only the second is a W_J .

The finite-sieve dictionary. To compute with sheaf samenesses on the simplex and cube categories we use an explicit dictionary (Theorem 5.1): on

a category with finitely many sieves on each object, Grothendieck topologies correspond, by reverse inclusion, to *idempotent two-sided ideals*, the ideal of a topology being the family of its least covering sieves. This is the finite-sieve form of the correspondence of Kelly and Lawvere [13] between essential localizations of a presheaf topos and idempotent ideals, and it is closely related to the rigidity results of Johnstone [5, C2.2.21], Lindenhovius [15] and Marquès [16]; we include the short direct proof because it is elementary, needs no Cauchy completeness, and gives the explicit least-cover description used later. Its consequences — a finite poset has $2^{|P|}$ topologies, a group two, a finite meet-semilattice L as a monoid has $\text{Top}(L) \cong \text{Down}(L^{\text{op}})$ while $\text{Same}(L) \cong L^{\text{op}}$, and the cube and simplex categories have exactly the dimension chain of topologies, so that their subtoposes are the coskeletal ones — are re-derivations of, or small additions to, known facts: that the essential subtoposes (levels) of simplicial and cubical sets are the dimensions is due to Kennett, Riehl, Roy and Zaks [14], and that every topology on Δ is rigid is in Marquès [16]. What we add is the position of these topologies as the sheaf chain $W_{D_0} \supseteq W_{D_1} \supseteq \dots$ in $\text{Same}(\mathbf{sSet})$, which is what the main theorem joins with W_{test} .

Verification. Every finite assertion (numbers of topologies, the bijection with idempotent ideals, the poset, monoid, semilattice and cube computations) was verified by enumeration from the definition of a Grothendieck topology (§8).

Related work. Subtoposes of presheaf toposes and their correspondence with Grothendieck topologies, local isomorphisms and left-exact reflections are classical [4, 5]; subtoposes of $\text{Sh}(X)$ for a locale X correspond to nuclei [6]. Kelly and Lawvere [13] showed that the essential localizations of a topos form a complete lattice and described those of a presheaf topos $\text{Presh}(S)$ through idempotent two-sided ideals of S ; Kennett, Riehl, Roy and Zaks [14] computed the resulting levels of simplicial, cubical and reflexive globular sets and their *Aufhebung*; Marquès [16] characterised, modulo slices, the categories on which every Grothendieck topology is rigid, which include finite Cauchy-complete categories, Artinian posets [15] and Δ . Toposes of monoid actions are studied by Hemelaer and Rogers [17, 18]. Coskeleta, Kan’s Ex^∞ and Postnikov sections are in [7]; that the coskeleta of a Kan complex model its Postnikov sections is folklore, and we include the short proof (Lemma 7.1). Composites and comparisons of localizations are a classical theme of the theory of Bousfield localization and homotopy idempotent functors [8, 11], where localizations are typically compared within model or homotopical structures and with additional localization or saturation structure; the join theorem (Theorem 4.1) and its corollaries assume none of this — no model structure, no factorization, no saturation — and hold in the complete lattice of bare two-out-of-three classes of an arbitrary category, which is what allows the sheaf-theoretic class $W_{D_{n+1}}$ to be joined with W_{test}

without first realising both as weak equivalences of model structures. Weak-equivalence classes, localizers and model structures on presheaf categories are studied by Cisinski and others; what is new here is the treatment of the bare two-out-of-three classes as a lattice in which sheaf and homotopy localizations can be joined, and the resulting identification of the Postnikov truncations as such joins.

2. PRELIMINARIES

2.1. Samenesses. For a small category C , a *sameness* is a class $W \subseteq \text{Mor}(C)$ with (i) all identities in W , (ii) W closed under composition, (iii) two-out-of-three: for composable f, g , if two of $f, g, g \circ f$ lie in W then so does the third. $\text{Same}(C)$ is the set of samenesses ordered by inclusion.

Proposition 2.1. *$\text{Same}(C)$ is a complete lattice (meets are intersections; $\perp = \{\text{id}\}$, $\top = \text{Mor}(C)$). For a functor $F: C \rightarrow D$ and $W \in \text{Same}(D)$, $F^*(W) = \{m : F(m) \in W\}$ is a sameness in C , and F^* preserves meets. In particular $V_F := F^*(\text{Iso}D)$ is the largest sameness inverted by F ; F factors through the localization $C \rightarrow C[V_F^{-1}]$, and the induced functor $C[V_F^{-1}] \rightarrow D$ is an equivalence iff F exhibits D as the localization of C at V_F . For a group G (as a one-object category), $\text{Same}(G)$ is the subgroup lattice $\text{Sub}(G)$.*

Proof. The three conditions are preserved by intersections and hold for $\text{Mor}(C)$; F^* preserves them because F preserves identities and composites, and it preserves intersections. The universal property of the localization [12, I.1] gives the factorization, and the last assertion is then the definition of “exhibits as the localization”. For a group, two-out-of-three applied to (w^{-1}, w) , whose composite is the identity, forces $w^{-1} \in W$, so a sameness is a subgroup; conversely every subgroup satisfies the three conditions. $\square$

Definition 2.2. For $W \in \text{Same}(C)$, an object Z is *W -local* if $\text{Hom}(f, Z)$ is bijective for every $f \in W$; write $\text{Loc}(W)$ for the class of W -local objects. The *saturation* of W is $\text{Sat}(W) := \{f : \text{Hom}(f, Z) \text{ bijective for all } Z \in \text{Loc}(W)\}$, a sameness containing W ; W is *saturated* if $W = \text{Sat}(W)$.

Lemma 2.3. *For $U, V \in \text{Same}(C)$, $\text{Loc}(U \vee V) = \text{Loc}(U) \cap \text{Loc}(V)$; hence $\text{Sat}(U \vee V) = \text{Sat}(\text{Sat}(U) \vee \text{Sat}(V))$. If $L: C \rightarrow C$ is a reflector onto a full replete subcategory $\mathcal{E}$ (with unit η), then $W_L := L^*(\text{Iso})$ is saturated, $\text{Loc}(W_L) = \mathcal{E}$, and $C \rightarrow \mathcal{E}$ is the localization of C at W_L .*

Proof. Z is local for $U \vee V$ iff for both U and V : one direction is trivial, and conversely $\{f : \text{Hom}(f, Z) \text{ bijective}\}$ is a sameness containing U and V , hence the join. The second assertion follows since $\text{Loc}(\text{Sat}(U)) = \text{Loc}(U)$. For a reflector, f is inverted by L iff $\text{Hom}(Lf, Z)$ is bijective for all $Z \in \mathcal{E}$ (Yoneda in $\mathcal{E}$) iff $\text{Hom}(f, Z)$ is bijective for all $Z \in \mathcal{E}$; so $\mathcal{E} \subseteq \text{Loc}(W_L)$ and $W_L = \{f : \text{Hom}(f, Z) \text{ bij. } \forall Z \in \mathcal{E}\}$. If Z is W_L -local then $\eta_Z \in W_L$ gives $\text{Hom}(\eta_Z, Z)$ bijective, so η_Z has a retraction r with $r\eta_Z = \text{id}$; then $\eta_Z r$ and id_{LZ} have the same composite with η_Z , and $\text{Hom}(\eta_Z, LZ)$ is bijective, so η_Z

is an isomorphism and $Z \in \mathcal{E}$. Hence $\text{Loc}(W_L) = \mathcal{E}$ and W_L is saturated. The localization statement is [12, I.1.3]. $\square$

2.2. Sieves, topologies, sheafification. A *sieve* on an object c of S is a set of morphisms with codomain c closed under precomposition; for $h: d \rightarrow c$ and a sieve R on c , $h^*R = \{g: h \circ g \in R\}$. A *Grothendieck topology* J assigns to each c a set $J(c)$ of sieves such that the maximal sieve is in $J(c)$, J is stable ($R \in J(c) \Rightarrow h^*R \in J(d)$) and transitive (if $R \in J(c)$ and Q is a sieve on c with $h^*Q \in J(d)$ for all $h: d \rightarrow c$ in R , then $Q \in J(c)$) [4]. Each $J(c)$ is then upward closed and closed under finite intersection. $\text{Top}(S)$ is the set of topologies ordered by $J \leq J'$ iff $J(c) \subseteq J'(c)$ for all c ; it is a complete lattice. We write $a_J: \text{Presh}(S) \rightarrow \text{Sh}_J(S)$ for sheafification, a left-exact left adjoint to the inclusion; $J \leq J'$ implies $a_{J'} = a_{J'} \circ a_J$ up to isomorphism. A morphism of presheaves is a *J -local isomorphism* if it is J -locally injective and J -locally surjective; $a_J f$ is an isomorphism iff f is a J -local isomorphism [4, III.7], [5, C2.2].

3. TOPOLOGIES EMBED IN THE SAMENESS LATTICE

Theorem 3.1. *Let S be a small category. For $J \in \text{Top}(S)$ put $W_J := \{f \in \text{MorPresh}(S) : a_J f \text{ is an isomorphism}\}$. Then:*

- (1) $W_J \in \text{Same}(\text{Presh}(S))$, and W_J is the class of J -local isomorphisms; it is stable under pullback and closed under colimits in the arrow category;
- (2) $J \mapsto W_J$ is an order-embedding $\text{Top}(S) \hookrightarrow \text{Same}(\text{Presh}(S))$;
- (3) $W_J = V_{a_J}$ is the largest sameness inverted by a_J , and the induced $\text{Presh}(S)[W_J^{-1}] \rightarrow \text{Sh}_J(S)$ is an equivalence;
- (4) the image of the embedding is the set of saturated samenesses W that are the full class of maps inverted by a left-exact reflector onto a full replete subcategory of $\text{Presh}(S)$; equivalently, $\text{Loc}(W)$ is a subtopos and $W = \text{Sat}(W)$.

Proof. (1) $W_J = a_J^*(\text{Iso})$ is a sameness by Proposition 2.1; the local isomorphism description is the classical characterisation of the maps inverted by sheafification; pullback stability because a_J is left exact, colimit closure because it is a left adjoint. (2) Monotone: $J \leq J'$ gives $a_{J'} = a_{J'} \circ a_J$, so $a_J f$ iso implies $a_{J'} f$ iso. Injective and order-reflecting: the W_J -local objects are exactly the J -sheaves (for a reflective localization the local objects form the reflective subcategory), and a subtopos of $\text{Presh}(S)$ determines its topology [4, V.4], [5, A4.3], so $W_J \subseteq W_{J'}$ forces $\text{Sh}_{J'} \subseteq \text{Sh}_J$, i.e. $J \leq J'$. (3) By Lemma 2.3, since a_J is a reflector. (4) If $W = L^*(\text{Iso})$ for a left-exact reflector L onto $\mathcal{E} \subseteq \text{Presh}(S)$, then $\mathcal{E}$ is a subtopos, hence $\mathcal{E} = \text{Sh}_J(S)$ for a unique J [5, A4.3], and $W = \{f : \text{Hom}(f, Z) \text{ bij. } \forall Z \in \mathcal{E}\} = W_J$ by Lemma 2.3; conversely $W_J = a_J^*(\text{Iso})$ is of this form and saturated. The condition “saturated” cannot be dropped: $\{\text{id}\}$ and Iso have the same localization but only $\text{Iso} = W_{J_{\min}}$ (trivial topology) is in the image. $\square$

Remark 3.2. $\text{Presh}(S)$ is large; as in [1] we fix universes $U \in V$, take S U -small and form Same and the localizations in V . Nothing below depends on this bookkeeping. Figure 1 summarises Theorem 3.1.

$$\begin{array}{ccc}
 \text{Top}(S) & \xrightarrow{J \mapsto W_J} & \text{Same}(\text{Presh}(S)) \\
 \downarrow J \mapsto \text{Sh}_J(S) & & \downarrow W \mapsto \text{Presh}(S)[W^{-1}] \\
 \{\text{subtoposes}\}^{\text{op}} & \hookrightarrow & \{\text{localizations}\} \\
 \text{Presh}(S) & \xrightarrow{a_J} & \text{Sh}_J(S) \\
 \downarrow \text{loc.} & \nearrow \simeq & \\
 \text{Presh}(S)[W_J^{-1}] & &
 \end{array}$$

FIGURE 1. Top: the order-embedding of Theorem 3.1 over the classical correspondence between topologies and subtoposes; the right-hand column sends a sameness to its localization, and the image of the top row is the set of saturated samenesses inverted by a left-exact reflector. Bottom: W_J is the largest sameness inverted by a_J , and the induced comparison is an equivalence.

Alt text: Above, a commutative square: the lattice of topologies embeds into the sameness lattice along the top, and both map down to subtoposes and localizations respectively. Below, a triangle: sheafification from presheaves to sheaves factors through the localization at the sameness of the topology, and the comparison functor is an equivalence.

4. JOINS OF SAMENESSES AND LOCALIZATIONS

The main theorem rests on the following formal statement.

Theorem 4.1 (join theorem). *Let $U, V, W \in \text{Same}(C)$ with $W \subseteq U \vee V$, and let $P: C \rightarrow C$ be a functor with a natural transformation $\pi: \text{id}_C \Rightarrow P$ such that*

- (a) $\pi_X \in U \vee V$ for every object X ;
- (b) $P(U) \subseteq W$ and $P(V) \subseteq W$.

Then $U \vee V = P^(W) = \{f : Pf \in W\}$.*

Proof. $P^*(W)$ is a sameness (Proposition 2.1) containing U and V by (b), hence containing $U \vee V$. Conversely let $f: X \rightarrow Y$ with $Pf \in W \subseteq U \vee V$. Naturality gives $\pi_Y \circ f = Pf \circ \pi_X$; the right-hand side lies in $U \vee V$ by (a), and $\pi_Y \in U \vee V$, so two-out-of-three gives $f \in U \vee V$. $\square$

Corollary 4.2 (two reflectors). *Let L_1, L_2 be reflectors of C onto full replete subcategories, with $W_{L_i} = L_i^*(\text{Iso})$, and write $W_{L_2 L_1} := (L_2 L_1)^*(\text{Iso})$ for the class of maps inverted by the composite functor (which need not itself be a reflector). Then*

$$W_{L_1} \vee W_{L_2} = W_{L_2 L_1} \iff L_2 L_1 \text{ inverts } W_{L_2}.$$

In particular the equality holds whenever L_1 preserves L_2 -equivalences, or whenever $L_2 L_1 \cong L_2$ (e.g. $\text{Loc}(W_{L_2}) \subseteq \text{Loc}(W_{L_1})$).

Proof. Apply Theorem 4.1 with $U = W_{L_1}$, $V = W_{L_2}$, $W = \text{Iso}$, $P = L_2 L_1$ and $\pi_X = \eta_{L_1 X}^2 \circ \eta_X^1$, a composite of a W_{L_1} -map and a W_{L_2} -map; (b) for U holds because $L_2 L_1$ inverts W_{L_1} , and (b) for V is the stated condition. Conversely, if the equality holds then $W_{L_2} \subseteq W_{L_2 L_1}$. $\square$

Proposition 4.3 (joins of topologies). *For Grothendieck topologies J, J' on S , with $J \vee J'$ their join in $\text{Top}(S)$,*

$$W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'}),$$

so $W_{J \vee J'} = W_J \vee W_{J'}$ iff the join $W_J \vee W_{J'}$ is saturated.

Proof. A presheaf is a $(J \vee J')$ -sheaf iff it is a J -sheaf and a J' -sheaf, because the sieves for which a given presheaf (and all its pullbacks) satisfies the sheaf condition form a topology [5, C2.1]. By Lemma 2.3, $\text{Loc}(W_J \vee W_{J'}) = \text{Sh}_J \cap \text{Sh}_{J'} = \text{Sh}_{J \vee J'} = \text{Loc}(W_{J \vee J'})$, and $W_{J \vee J'}$ is saturated, so $W_{J \vee J'} = \text{Sat}(W_J \vee W_{J'})$. $\square$

Example 4.4. Let S be the poset $0 < 1$, so that $\text{Presh}(S)$ is the category of maps of sets $X(1) \rightarrow X(0)$. The two intermediate topologies (§5) have sheaf samenesses $W_0 = \{f : f_0 \text{ bijective}\}$ and $W_1 = \{f : f_1 \text{ bijective}\}$, and their join topology is the degenerate one, with $W = \text{Mor}$. Here $W_0 \vee W_1 = \text{Mor}$: any $f = (f_1, f_0) : X \rightarrow Y$ factors as $(X(1) \rightarrow X(0)) \xrightarrow{(\text{id}, f_0)} (X(1) \rightarrow Y(0)) \xrightarrow{(f_1, \text{id})} (Y(1) \rightarrow Y(0))$, a W_1 -map followed by a W_0 -map. Whether $W_J \vee W_{J'} = W_{J \vee J'}$ holds for all pairs of topologies on all sites we leave open; Proposition 4.3 reduces it to the saturation of the join.

5. THE FINITE-SIEVE DICTIONARY

A *two-sided ideal* of S is a class D of morphisms closed under pre- and post-composition with arbitrary morphisms of S ; write $D(c)$ for its members with codomain c (a sieve on c). D is *idempotent* if $D \circ D = D$, i.e. every member of D is a composite of two members of D . Let $\text{Idl}_{\text{idem}}^2(S)$ be the set of idempotent two-sided ideals.

Kelly and Lawvere [13] describe the essential localizations of $\text{Presh}(S)$ — those whose reflector has a left adjoint — in terms of idempotent two-sided ideals of S ; when every object has finitely many sieves, every topology has least covering sieves and the description becomes the following elementary dictionary, which we prove directly since it is short, requires no Cauchy

completeness, and is what we compute with. (For categories with finite slices [5, C2.2.21], [13, Prop. 4.10], and for those in which every arrow factors as a split epimorphism followed by a split monomorphism and every object has finitely many subobjects (Menni; see criterion (C2) in [16]), every topology is moreover *rigid*: its sheaf topos is a presheaf topos on the full subcategory of J -irreducible objects, i.e. the objects c with $\text{id}_c \in D_J$ below.)

Theorem 5.1 (ideal theorem). *Let S have finitely many sieves on each object (e.g. S finite). Then*

$$\text{Top}(S) \cong \text{Idl}_{\text{idem}}^2(S)^{\text{op}},$$

$$J \mapsto D_J := \left(\bigcap_c J(c) \right)_c, \quad D \mapsto J_D(c) := \{R \mid R \text{ a sieve on } c : R \supseteq D(c)\},$$

mutually inverse order-reversing bijections. The trivial topology corresponds to $D = \text{Mor}(S)$ and the degenerate one (all sieves) to $D = \emptyset$.

Proof. Let J be a topology. Each $J(c)$ is upward closed and closed under finite intersection, so with finitely many sieves $D_J(c) := \bigcap J(c) \in J(c)$ and $J(c) = \{R \supseteq D_J(c)\}$. Stability of $D_J(c)$ along $h: d \rightarrow c$ gives $h^*D_J(c) \in J(d)$, hence $h^*D_J(c) \supseteq D_J(d)$, i.e. $h \circ g \in D_J(c)$ for every $g \in D_J(d)$: D_J is closed under post-composition, and being a family of sieves it is closed under pre-composition, so it is a two-sided ideal. Transitivity applied to $R = D_J(c)$ and $Q = (D_J \circ D_J)(c)$: for $h \in D_J(c)$, $h^*Q \supseteq D_J(d)$ since $h \circ g \in D_J \circ D_J$ for $g \in D_J(d)$; hence $Q \in J(c)$, so $Q \supseteq D_J(c)$, i.e. $D_J \circ D_J \supseteq D_J$; the other inclusion is trivial, so D_J is idempotent.

Conversely let D be an idempotent two-sided ideal. $J_D(c)$ contains the maximal sieve. Stability: if $R \supseteq D(c)$ and $g \in D(d)$ then $h \circ g \in D(c) \subseteq R$, so $h^*R \supseteq D(d)$. Transitivity: let $R \supseteq D(c)$ and Q a sieve with $h^*Q \supseteq D(d)$ for all $h: d \rightarrow c$ in R ; in particular for $h \in D(c)$, so $h \circ D(d) \subseteq Q$ for all $h \in D(c)$, i.e. $(D \circ D)(c) \subseteq Q$, whence $D(c) \subseteq Q$ by idempotence and $Q \in J_D(c)$. The two assignments are inverse ($D_{J_D} = D$ since $D(c)$ is the least sieve containing $D(c)$; $J_{D_J} = J$ as shown) and reverse inclusion. $\square$

The theorem was checked by enumerating topologies from the definition and idempotent two-sided ideals independently, on sixty-one monoids (all of order ≤ 4 , cyclic groups, thirteen lattices as meet-monoids) and on ten categories with several objects, parallel arrows and non-trivial endomorphisms; the bijection and its order-reversal held in every case (§8); see Figure 2.

Corollary 5.2 (posets). *For a finite poset P (as a category), $\text{Top}(P) \cong 2^{|P|}$, the Boolean lattice on the underlying set: the topologies depend only on the number of points, not on the order. (This also follows from rigidity: the topologies on an Artinian poset correspond to its subsets [15], [16].)*

Proof. For $Y \subseteq P$ let D_Y be the set of arrows $a \rightarrow b$ with $a \leq y \leq b$ for some $y \in Y$; it is a two-sided ideal, idempotent since $a \rightarrow b$ factors as $(y \rightarrow b) \circ (a \rightarrow y)$ with both factors in D_Y , and Y is recovered from D_Y as the set of objects whose identity it contains. Conversely let D be an

$$\begin{array}{ccc}
 J & \xrightarrow{D_J = (\bigcap J(c))_c} & D_J \\
 J_{D_J} = J \parallel & & \parallel D_{J_D} = D \\
 J_D & \xleftarrow{J_D(c) = \{R \supseteq D(c)\}} & D
 \end{array}$$

FIGURE 2. The ideal theorem (Theorem 5.1): a topology is recorded by its least covering sieves, which form an idempotent two-sided ideal, and an idempotent two-sided ideal generates a topology by upward closure; the two passages are inverse and reverse the order.

Alt text: A square of assignments. A topology J maps right to the ideal of its least covering sieves; an ideal D maps left to the topology of sieves containing D ; the vertical equalities record that the two round trips are identities.

idempotent two-sided ideal and Y the set of objects with $\text{id}_y \in D$; then $D \supseteq D_Y$. For $a \rightarrow b$ in D , idempotence gives y_1 with $a \leq y_1 \leq b$ and $a \rightarrow y_1, y_1 \rightarrow b$ in D ; if $y_1 = a$ or $y_1 = b$ an identity of D lies between a and b ; otherwise repeat with $a \rightarrow y_1$. Since P is finite the strictly shrinking intervals terminate, producing $y \in Y$ with $a \leq y \leq b$, so $a \rightarrow b \in D_Y$. Hence $D = D_Y$, and $Y \mapsto D_Y$ is an order-isomorphism $2^P \cong \text{Idl}_{\text{idem}}^2(P)$. Verified from the definition of a topology on eight posets with ≤ 4 points (chains, antichains, V , Λ , the square), in each case also matching the number of nuclei on the frame $\text{Down}(P)$. $\square$

Corollary 5.3 (groups, groupoids, monoids). *For a monoid M with finitely many right ideals, $\text{Top}(M)$ is the poset of idempotent two-sided ideals of M under reverse inclusion. For a group (or a connected groupoid) the only two-sided ideals are $\emptyset$ and everything, so $\text{Top}(G)$ is the two-element chain, while $\text{Same}(G) = \text{Sub}(G)$; for a groupoid with a set π_0 of components, $\text{Top} \cong 2^{\pi_0}$.*

Corollary 5.4 (semilattices). *Let L be a finite meet-semilattice with top element, viewed as a monoid under $\wedge$. Then every ideal (down-set) of L is a two-sided idempotent ideal, so*

$$\text{Top}(L) \cong \text{Down}(L)^{\text{op}} \cong \text{Down}(L^{\text{op}}),$$

the lattice of up-sets of L ; in particular the lattice of subtoposes of the topos of L -sets determines L up to isomorphism (Birkhoff duality). The arrow-level sameness lattice is $\text{Same}(L) \cong L^{\text{op}}$ via $a \mapsto \uparrow a$; hence $\text{Top}(L) \cong \text{Down}(\text{Same}(L))$.

Proof. A down-set D satisfies $D \wedge L \subseteq D$ and $d = d \wedge d$, so it is a two-sided idempotent ideal; conversely a right ideal is a down-set. For $\text{Same}(L)$: a sameness W is a meet-closed submonoid; with $a := \bigwedge W \in W$ we have $W \subseteq \uparrow a$, and for $y \geq a$ the pair (y, a) has composite $y \wedge a = a \in W$, so

two-out-of-three gives $y \in W$; each $\uparrow a$ is a sameness, and $\uparrow a \subseteq \uparrow b$ iff $b \leq a$. Verified on thirteen lattices (n -chains, 2^2 , 2^3 , M_3 , N_5 , 2×3 , $1 \oplus 2^2$, $2^2 \oplus 1$, $2^2 \oplus 2$): e.g. $|\text{Top}(2^3)| = 20$, $|\text{Top}(M_3)| = 10$, $|\text{Top}(N_5)| = 8$. $\square$

Example 5.5. Among the 35 monoids of order 4, $\text{Top}(M)$ is a chain for 34; the unique exception is the meet-semilattice 2^2 (identity, zero, two orthogonal idempotents), with $\text{Top}(2^2) \cong \text{Down}(2^2)^{\text{op}}$, six elements (Figure 3). The pairs $(|\text{Top}M|, |\text{Same}M|)$ take nine different values on order 4; neither invariant determines the other.

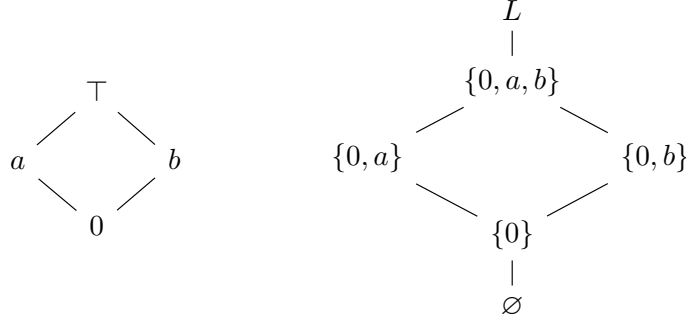


FIGURE 3. Left: the meet-semilattice $L = 2^2$ as a monoid. Right: its six topologies, $\text{Top}(L) \cong \text{Down}(L)^{\text{op}}$ (drawn as $\text{Down}(L)$; the topology order is the reverse). The arrow-level lattice is $\text{Same}(L) \cong L^{\text{op}}$, four elements.

Alt text: Two Hasse diagrams. Left, a diamond with bottom 0, two incomparable elements a and b , and top. Right, a six-element lattice: empty set at the bottom, then the singleton of 0, then two incomparable sets $\{0, a\}$ and $\{0, b\}$, then $\{0, a, b\}$, then the whole semilattice at the top.

6. CUBE AND SIMPLEX CATEGORIES

Let $\square$ be the cube category generated by faces and degeneracies (Kan's cube category, as in [14]; no connections, reversals or symmetries) and Δ the simplex category. For $k \geq 0$ let D_k be the class of morphisms that factor through an object of dimension $\leq k$. The following theorem is a re-derivation, through the dictionary, of known facts: that the levels (essential subtoposes) of simplicial and cubical sets are the dimensions is [14, §1], and that every topology on Δ is rigid is [16, Thm. 2.6], so that all subtoposes of $\mathbf{sSet}$ are levels; the same holds for $\square$, which satisfies the split-epi/split-mono factorization criterion of Menni recorded as (C2) in [16]. What we need from it is the last sentence: the explicit description of the sheaf samenesses.

Theorem 6.1. *The two-sided ideals of $\square$ (resp. Δ) are exactly $\emptyset$, the D_k ($k \geq 0$) and Mor ; all are idempotent. Hence the topologies form the chain*

$$\text{degenerate} \subset \cdots \subset J_{D_1} \subset J_{D_0} \subset \text{trivial}$$

(reverse inclusion of ideals), and the J_{D_k} -sheaves are the k -coskeletal presheaves: the subtoposes of cubical sets and of simplicial sets are precisely the coskeletal ones. The sheaf samenesses form the descending chain $W_{D_0} \supseteq W_{D_1} \supseteq \cdots$ of k -coskeletal equivalences,

$$W_{D_k} = \{f : \text{cosk}_k f \text{ iso}\} = \{f : f_m \text{ bijective for all } m \leq k\},$$

with $\bigcap_k W_{D_k} = \text{Iso}$.

Proof. Every morphism $f : [m] \rightarrow [n]$ factors as a degeneracy $\sigma : [m] \twoheadrightarrow [k]$ followed by a face $\delta : [k] \hookrightarrow [n]$, k the rank of f ; degeneracies have sections and faces have retractions among the generators. So $\text{id}_{[k]} = r \circ f \circ s$ lies in the two-sided ideal generated by f , which therefore contains every morphism factoring through $[k]$, i.e. D_k , and $f \in D_k$. Hence principal ideals are the D_k , two-sided ideals are unions of D_k 's — $\emptyset$, some D_k , or everything — and each is idempotent since it contains an identity through which its members factor. Theorem 5.1 gives the topologies: its finiteness hypothesis holds, a sieve on $[n]$ being a subpresheaf of the representable $y[n]$, which is determined by the finitely many non-degenerate cells it contains (every cell of $y[n]$ is uniquely a degeneracy of a non-degenerate one). The least covering sieve of J_{D_k} on $[n]$ is $D_k([n])$, generated by the maps from dimension $\leq k$, so a sheaf X satisfies $X([n]) \cong \text{Hom}(\text{sk}_k y[n], X)$ for all n , i.e. $X \cong \text{cosk}_k X$; conversely a k -coskeletal X is local for every sieve $R \supseteq \text{sk}_k y[n]$, since $\text{Hom}(R, \text{cosk}_k X) = \text{Hom}(\text{sk}_k R, X)$ and $\text{sk}_k R = \text{sk}_k y[n]$. Since $(\text{cosk}_k X)_m = \text{Hom}(\text{sk}_k y[m], X)$ depends only on the k -truncation of X and equals X_m for $m \leq k$, $\text{cosk}_k f$ is an isomorphism iff f is bijective in dimensions $\leq k$; the intersection is therefore Iso . $\square$

Verified on the truncations $\square_{\leq 1}$ (7 morphisms), $\square_{\leq 2}$ (37), $\Delta_{\leq 1}$ (7), $\Delta_{\leq 2}$ (31): exactly $\emptyset, D_0, \dots, D_N$ in each case. Figure 4 shows the chain together with its sheaves.

7. SHEAF SAMENESS VERSUS HOMOTOPY SAMENESS

On $\mathbf{sSet} = \text{Presh}(\Delta)$ the lattice $\text{Same}(\mathbf{sSet})$ contains the homotopy sameness W_{test} (weak homotopy equivalences) and the sheaf chain W_{D_k} . Let $\text{Ex}^\infty : \mathbf{sSet} \rightarrow \mathbf{sSet}$ be Kan's fibrant replacement, with its natural weak equivalence $X \rightarrow \text{Ex}^\infty X$ into a Kan complex [7, III.4]. For $n \geq -1$ put

$$P_n := \text{cosk}_{n+1} \circ \text{Ex}^\infty : \mathbf{sSet} \rightarrow \mathbf{sSet},$$

$$T_n := P_n^*(W_{\text{test}}) = \{f : P_n f \text{ is a weak equivalence}\}.$$

By Proposition 2.1, T_n is a sameness. The following lemma is standard: the coskeleta of a Kan complex model its Postnikov tower (Duskin [9]; cf. the Postnikov towers of [7, VI]). We include the elementary proof for completeness, since the main theorem rests on it.

Lemma 7.1. *Let X be a Kan complex and $n \geq -1$, and put $K = \text{cosk}_{n+1} X$. Then K is a Kan complex; the unit $\eta : X \rightarrow K$ induces isomorphisms $\pi_i(X, x) \rightarrow \pi_i(K, x)$ for all $0 \leq i \leq n$ and all vertices x ; and $\pi_i(K, x) = 0$*

ideal D	topology J_D	sheaves	sameness W_D
$\emptyset$	degenerate	terminal only	Mor
D_0	J_{D_0}	0-coskeletal	W_{D_0}
D_1	J_{D_1}	1-coskeletal	W_{D_1}
$\vdots$	$\vdots$	$\vdots$	$\vdots$
Mor	trivial	all presheaves	Iso

FIGURE 4. Theorem 6.1: the four faces of the dimension chain for $\square$ and Δ . Ideals grow downwards, topologies and samenesses grow upwards; the sheaves for J_{D_k} are the k -coskeletal presheaves.

Alt text: A table of four columns aligned as a vertical chain: the ideals from empty to everything, the corresponding topologies from degenerate to trivial, their sheaves from the terminal object through the coskeletal presheaves to all presheaves, and the samenesses from all morphisms down to the isomorphisms.

for all $i \geq n + 1$ and all x (for $n = -1$: K is empty or contractible). Consequently, for $n \geq 0$,

$$T_n = \{f : \pi_0 f \text{ bijective, } \pi_i f \text{ iso at every base point, } 1 \leq i \leq n\},$$

and T_{-1} is the class of maps $X \rightarrow Y$ with $X = \emptyset$ iff $Y = \emptyset$.

Proof. An m -simplex of K is a map $\text{sk}_{n+1}\Delta^m \rightarrow X$, and η is bijective in dimensions $\leq n + 1$. *Kan condition.* A horn $\Lambda_m^k \rightarrow K$ is a map $\text{sk}_{n+1}\Lambda_m^k \rightarrow X$. If $m \geq n + 3$ then $\text{sk}_{n+1}\Lambda_m^k = \text{sk}_{n+1}\Delta^m$, so the horn is already an m -simplex of K . If $m \leq n + 2$ then Λ_m^k has dimension $\leq n + 1$, so the horn is a horn in X ; a filler $\Delta^m \rightarrow X$ restricts to $\text{sk}_{n+1}\Delta^m \rightarrow X$, an m -simplex of K extending it. *Homotopy groups.* For a Kan complex, $\pi_i(-, x)$ is computed from the i -simplices with boundary at x modulo the relation given by $(i+1)$ -simplices [7, I.7]; for $i \leq n$ these dimensions are $\leq n + 1$, where η is bijective, so $\pi_i \eta$ is an isomorphism. For $i \geq n + 1$, a map $\partial\Delta^{i+1} \rightarrow K$ is a map $\text{sk}_{n+1}\partial\Delta^{i+1} = \text{sk}_{n+1}\Delta^{i+1} \rightarrow X$, i.e. an $(i+1)$ -simplex of K extending it; a Kan complex with this extension property for all $i \geq n + 1$ has $\pi_i = 0$ for $i \geq n + 1$ at all base points [7, I.7]. *The description of T_n .* A map of Kan complexes is a weak equivalence iff it induces isomorphisms on π_0 and on all π_i at all base points, and Ex^∞ preserves homotopy groups [7, III.4]; apply this to $\text{cosk}_{n+1}\text{Ex}^\infty f$ using the two previous assertions. $\square$

Theorem 7.2 (join = truncation). *In $\text{Same}(\mathbf{sSet})$, for every $n \geq -1$,*

$$W_{\text{test}} \vee W_{D_{n+1}} = T_n.$$

Proof. We apply Theorem 4.1 with $U = W_{\text{test}}$, $V = W_{D_{n+1}}$, $W = W_{\text{test}}$ and $P = P_n$, so that $P^*(W) = T_n$ by definition. The natural transformation $\pi_X: X \rightarrow \text{Ex}^\infty X \rightarrow \text{cosk}_{n+1} \text{Ex}^\infty X$ is the Kan replacement, a weak equivalence, followed by the coskeleton unit, which is bijective in dimensions $\leq n+1$ (Theorem 6.1); so $\pi_X \in V \circ U \subseteq U \vee V$, which is (a). For (b): P_n preserves weak equivalences. Indeed Ex^∞ does; and if $g: X \rightarrow Y$ is a weak equivalence between Kan complexes, then by Lemma 7.1 the units $X \rightarrow \text{cosk}_{n+1} X$ and $Y \rightarrow \text{cosk}_{n+1} Y$ induce isomorphisms on π_i for $i \leq n$, so naturality shows that $\text{cosk}_{n+1} g$ induces isomorphisms on π_i for $i \leq n$ at all base points, while for $i > n$ both $\pi_i(\text{cosk}_{n+1} X)$ and $\pi_i(\text{cosk}_{n+1} Y)$ vanish; as $\text{cosk}_{n+1} X$ and $\text{cosk}_{n+1} Y$ are Kan, $\text{cosk}_{n+1} g$ is a weak equivalence. Thus $P_n(U) \subseteq W$. If $f \in W_{D_{n+1}}$ then f is bijective in dimensions $\leq n+1$, so $|\text{sk}_{n+1} X| \rightarrow |\text{sk}_{n+1} Y|$ is a homeomorphism. Now $|\text{sk}_{n+1} X|$ is the $(n+1)$ -skeleton of the CW complex $|X|$ (one cell per non-degenerate simplex), and the inclusion of the $(n+1)$ -skeleton of a CW complex induces isomorphisms on π_i for $0 \leq i \leq n$ at every base point (the pair $(|X|, |\text{sk}_{n+1} X|)$ is $(n+1)$ -connected [10, Cor. 4.12]). In the commutative square of $|\text{sk}_{n+1} f|$ and $|f|$ with these inclusions, three maps induce isomorphisms on π_i , $i \leq n$, hence so does $|f|$; and f preserves emptiness. Hence $f \in T_n$ by Lemma 7.1, i.e. $P_n f \in W$, which is $P_n(V) \subseteq W$. Figure 5 displays the naturality square used in the proof of Theorem 4.1 in this case. $\square$

So the Postnikov truncations — at the ∞ -level the n -truncation modalities — are exactly the joins, in the sameness lattice of simplicial sets, of the homotopy sameness with the sheaf samenesses of the coskeletal topologies; at $n = -1$ the join with the 0-coskeletal sameness is the (-1) -truncation, which identifies all non-empty simplicial sets. This is a comparison of two theories carried out entirely by a lattice operation, and it needs no fibrancy hypothesis: the Kan replacement is itself available inside W_{test} . Restricting along $\mathbf{Kan} \hookrightarrow \mathbf{sSet}$ (Proposition 2.1) the same identity holds in $\text{Same}(\mathbf{Kan})$.

Corollary 7.3 (sheaf-theoretic Whitehead). *In $\text{Same}(\mathbf{sSet})$,*

$$W_{\text{test}} = \bigcap_{n \geq 0} (W_{\text{test}} \vee W_{D_n}), \quad \text{Iso} = \bigcap_{n \geq 0} W_{D_n}.$$

Proof. By Theorem 7.2 the first intersection is $\bigcap_{n \geq -1} T_n$, the maps inducing isomorphisms on all homotopy groups at all base points, which is W_{test} by Whitehead's theorem (after Kan replacement). The second is Theorem 6.1. $\square$

Thus the homotopy sameness is the meet of its joins with the sheaf chain, while the sheaf chain itself meets down to the isomorphisms: the two families are “transversal” in $\text{Same}(\mathbf{sSet})$, and the truncations T_n are the interval they span (Figure 6).

$$\begin{array}{ccc}
X & \xrightarrow{f} & Y \\
\downarrow W_{\text{test}} & & \downarrow W_{\text{test}} \\
X' & \xrightarrow{f'} & Y' \\
\downarrow \eta_{X'} \quad W_{D_{n+1}} & & W_{D_{n+1}} \quad \downarrow \eta_{Y'} \\
\text{cosk}_{n+1} X' & \xrightarrow[\quad P_n f \quad]{\quad W_{\text{test}} \quad} & \text{cosk}_{n+1} Y'
\end{array}$$

FIGURE 5. Theorem 7.2 as an instance of Theorem 4.1: the vertical maps of the upper square are weak equivalences (Kan replacement), those of the lower square are coskeletal equivalences, and the bottom map is a weak equivalence when $f \in T_n$; two-out-of-three moves membership in the join from the bottom row to f .

Alt text: Two stacked commutative squares. Top row: f from X to Y . Middle row: the Kan replacement f' from X' to Y' , reached by vertical weak equivalences. Bottom row: the coskeleton of f' , labelled as a weak equivalence, reached by vertical coskeletal equivalences.

The analogous statement for cubical sets, where the n -coskeletality of [3] concerns uniformly fibrant objects, is the natural next step.

8. COMPUTATIONAL RECORD

All finite claims are verified from the definitions by the Python scripts supplied as supplementary material:

- `sheaf_sameness.py`: topologies on posets from the definition, nuclei on $\text{Down}(P)$ (Corollary 5.2);
- `sheaf_sameness_monoid.py`, `sheaf_sameness_monoid4.py`: monoids of order ≤ 3 and all 35 monoids of order 4;
- `sheaf_sameness_semilattice.py`: thirteen lattices as meet-monoids (Corollary 5.4);
- `sheaf_sameness_ideals.py`: Theorem 5.1 on 61 monoids;
`sheaf_sameness_general.py`: Theorem 5.1 on ten categories with several objects;
- `sheaf_sameness_cube.py`: Theorem 6.1 on the truncations $\square_{\leq 2}$ and $\Delta_{\leq 2}$.

Remark 8.1 (the sheaf level-shift logic). In [2] the modal logic of the frame $(\text{Same}(C), \subseteq)$ was studied. Restricting to the sheaf samenesses gives the modal logic of the lattice of subtoposes, which by Theorem 5.1 is, for a category with finitely many sieves per object, the logic of its poset of idempotent two-sided ideals: a combinatorial invariant of composition, computable without forming a presheaf. For $\square$ and Δ this frame is the chain of Theorem 6.1,

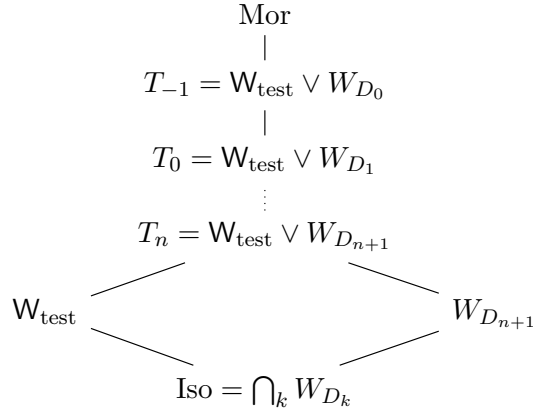


FIGURE 6. The position of the sheaf chain and the homotopy sameness in $\text{Same}(\mathbf{sSet})$ (Theorem 7.2 and Corollary 7.3): the joins along the chain are the Postnikov truncations, decreasing to W_{test} ; the chain itself decreases to Iso .

Alt text: A Hasse-style diagram. At the bottom the isomorphisms; above them, to the left the weak equivalences and to the right the coskeletal sameness of level $n + 1$; their join, the n -truncated equivalences, sits above both and is part of a chain of truncations rising to the (-1) -truncation and then to all morphisms.

so the logic contains S4.3; for a finite poset on n points it is the logic of the Boolean frame 2^n ; for every monoid of order ≤ 4 except the semilattice 2^2 it is the logic of a finite chain.

9. CONCLUSION

Sheaf theory’s notion of sameness is a family of saturated elements of the sameness lattice, intrinsically described (local isomorphisms), and within these elements the join of two topologies is the saturation of the join of the corresponding samenesses. On the cube and simplex categories the sheaf samenesses form the dimension chain of coskeletal equivalences, and on simplicial sets the join of this chain with the homotopy sameness is exactly the Postnikov truncation, for every $n \geq -1$, with the homotopy sameness recovered as the meet of these joins; the mechanism is a general join theorem for samenesses. Two questions are natural next steps: the cubical version of Theorem 7.2, where the n -coskeletality of [3] is precisely the sheaf sameness; and the ∞ -categorical statement, where reflective subuniverses (modalities) replace reflective samenesses and Theorem 7.2 should read “truncation = homotopy $\vee$ coskeleton” in the lattice of modalities.

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Availability of data and materials. All verification scripts named in §8 are provided as supplementary material.

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